

# Exploratory Data Analysis of the Greece airbnb market

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# Problem Statement

High-revenue listings are often assumed to be top performers. But revenue can simply reflect a higher price, not real guest demand. In this dataset, listings priced above \$500 earn roughly 3.5x more revenue than those under \$75, despite having about 4x lower occupancy. This project investigates whether elevated revenue actually reflects strong performance (occupancy), guest satisfaction, and host quality, or whether it's just a higher price tag. The goal is to help hosts and investors tell real market success apart from price inflation.



# Project Objectives



## **PRICE VS. OCCUPANCY AND REVENUE**

Determine whether higher-priced listings earn more due to genuine demand, or simply by charging more while occupying fewer nights.

## **RATING VS. OCCUPANCY AND REVENUE**

Assess whether listings with higher guest ratings achieve greater occupancy and revenue, or whether price is the dominant driver instead of rating.

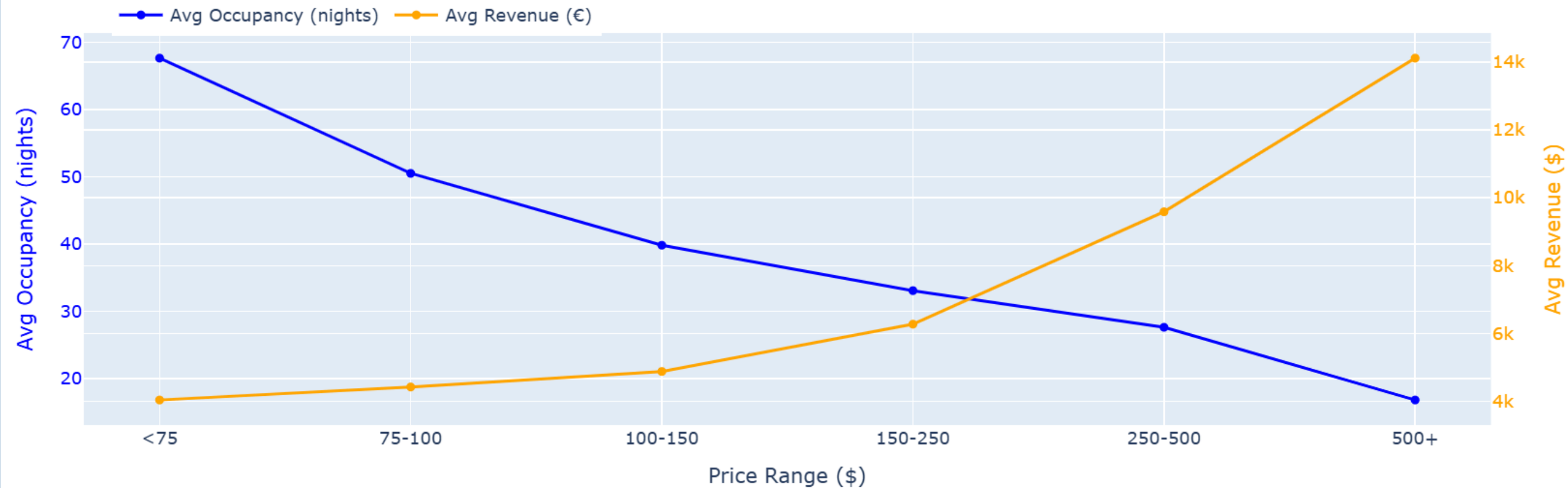
## **SUPERHOST VS. REGULAR HOST PERFORMANCE**

Evaluate whether Superhost listings genuinely outperform regular hosts in occupancy and revenue when controlling for price level, or whether the gap is simply a function of pricing.

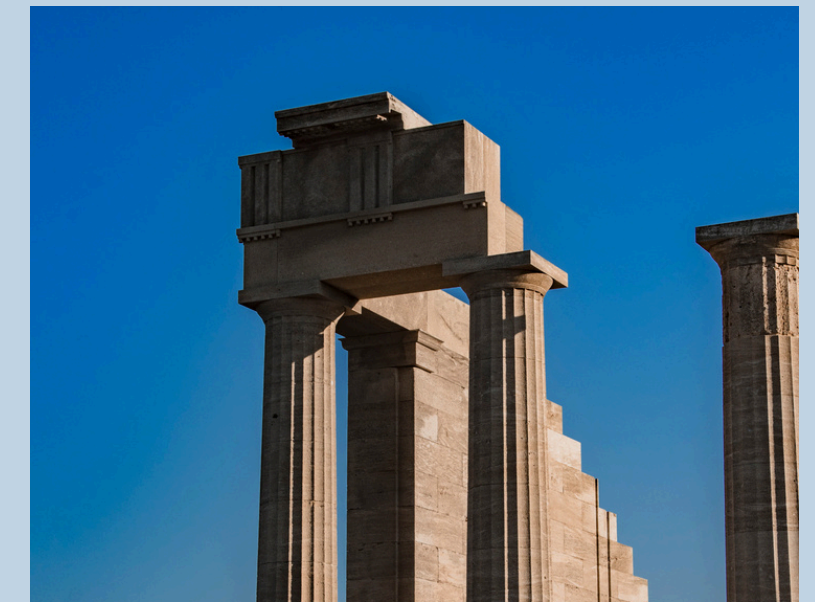
# Objective 1

## Price vs. Occupancy and Revenue

Objective 1: Price vs. Occupancy and Revenue



As price rises from under 75 \$ to over 500\$ , occupancy drops from ~68 to ~17 nights, yet revenue climbs from ~4k\$ to ~14k\$ — proving higher revenue comes from charging more, not from more bookings.

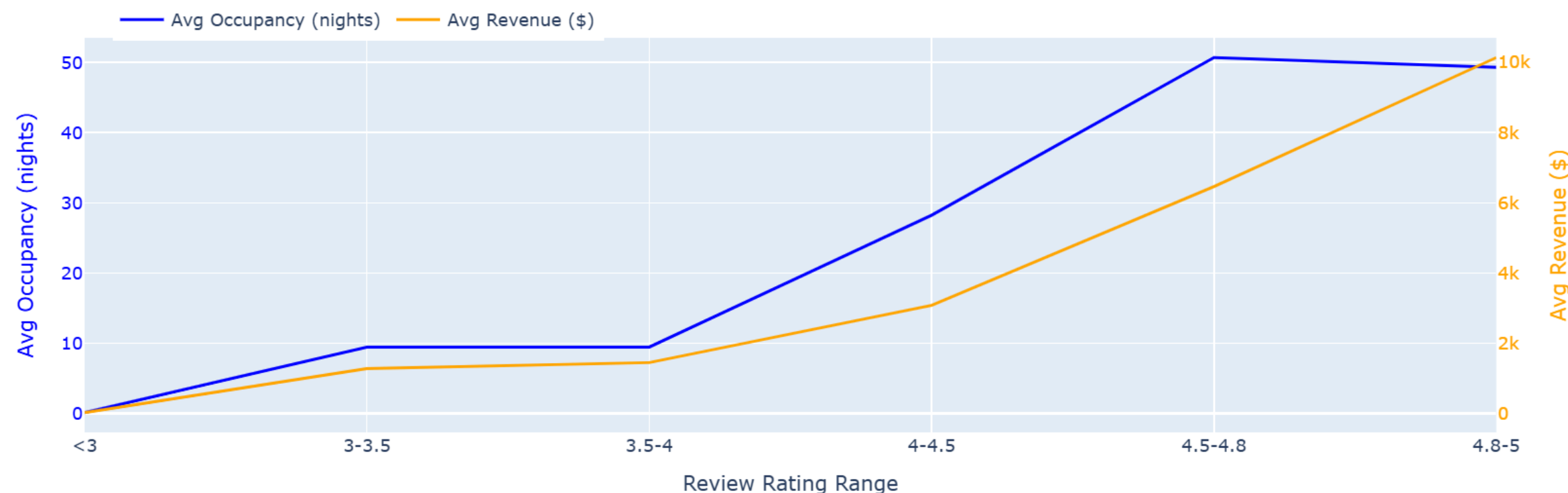


## Objective 2

# Rating vs. Occupancy and Revenue



Objective 2: Rating vs. Occupancy and Revenue



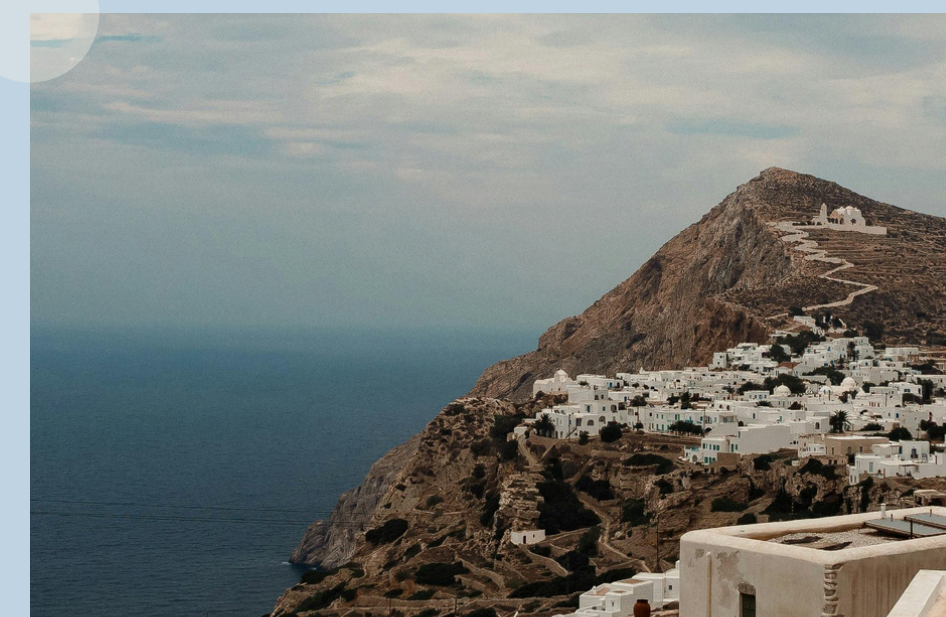
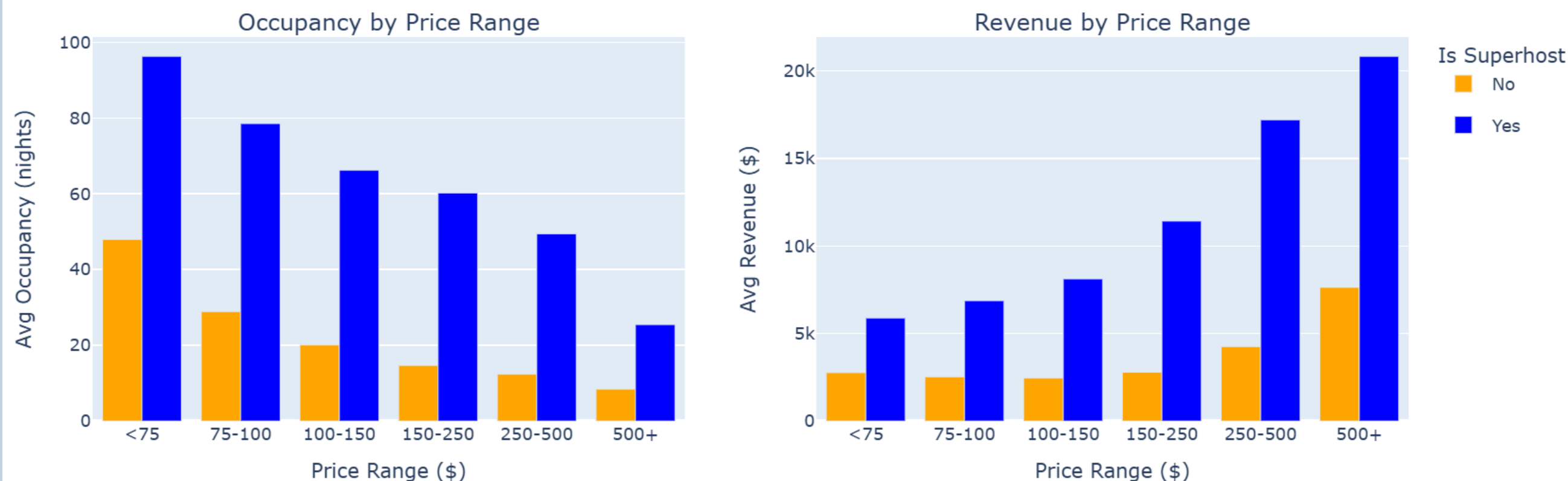
Listings rated below 3.5 average under 10 nights of occupancy and less than 1,500\$ in revenue, while listings rated 4.8–5 reach ~50 nights and over 10,000\$ showing rating drives real performance, not just price.

# Objective 3

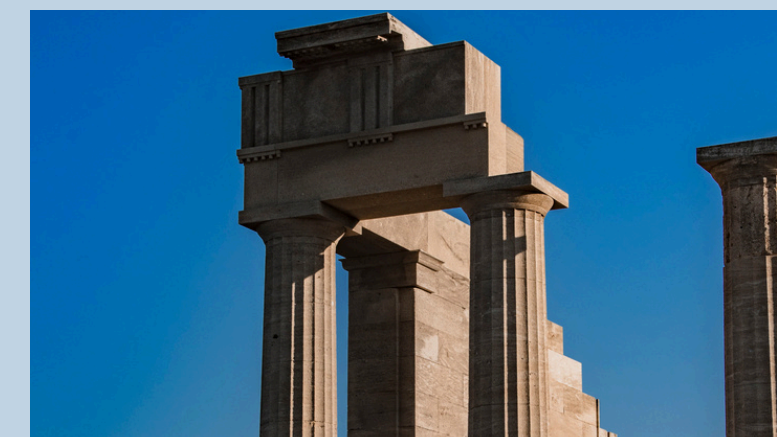
## Superhost vs. Regular Host Performance

At every price level, Superhosts show higher occupancy and revenue than regular hosts and the revenue gap widens sharply at higher prices (500\$+: ~21k\$ vs ~8k\$), proving the difference isn't just about price.

Objective 3: Occupancy & Revenue by Price Range — Superhost vs Regular



# Recommendations

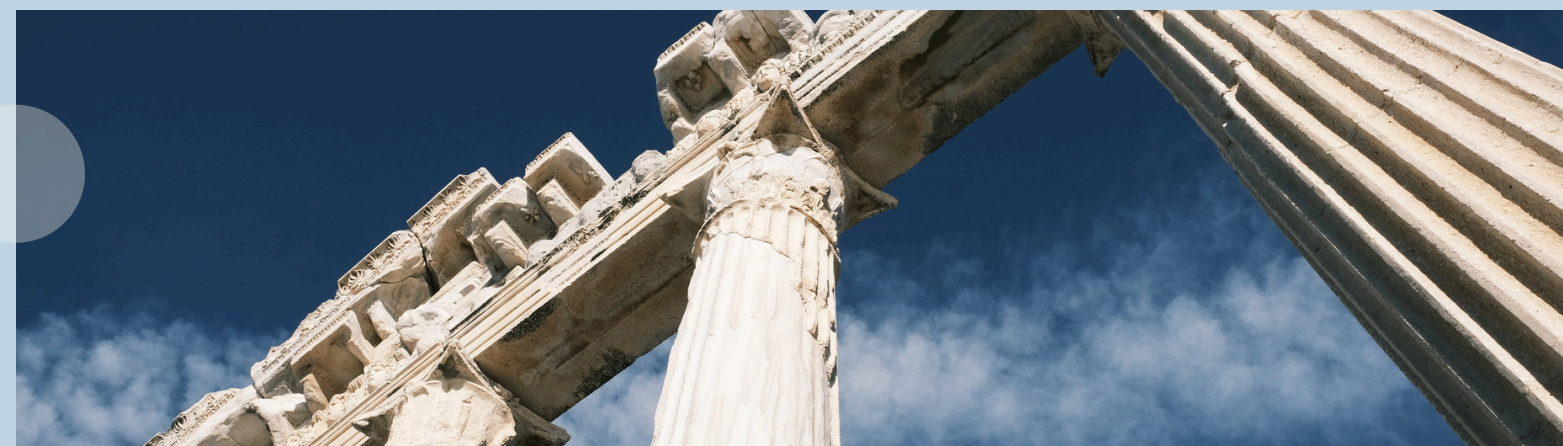


Evaluate listing performance using revenue-per-occupied-night rather than total revenue, to separate genuine demand from price inflation.

Prioritize Superhost standards (responsiveness, guest ratings) as an investment lever — their impact holds across all price tiers, independent of property size.

Weigh high-price, high-seasonality locations (islands) against lower-price, stable-demand locations (cities) based on annual revenue consistency, not peak pricing alone.

# Limitations



The dataset provides no criteria or timeline for how a listing attained Superhost status, so causality between the badge and performance cannot be established.

Approximately 20% of listings have no reviews, limiting the ability to assess their performance or include them in rating-based analysis.

The data represents a single point-in-time snapshot (29 June–3 July 2026); the analysis is descriptive (EDA) only and does not support causal claims or forecasting.

# Considerations

Findings reflect the Greek market only  
patterns may not generalize to other  
countries.

Price buckets and rating ranges are  
researcher-defined; different groupings  
could shift results slightly.

Correlation observed here doesn't confirm  
which factor changes first — rating,  
occupancy, and revenue likely influence  
each other.



# Thank You

